



# Procurement portfolio management for resilient supply chains: value of information for risk averse decision making

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## Abstract

In this study, we investigate the replenishment decisions for firms that rely on multiple sources of supplies, including regular forward contracts, option contracts, and the spot market. Our research aims to tackle the challenges arising due to supplier disruptions and the volatility of spot market prices that are correlated with these disruptions. We develop and solve multi-stage stochastic programming models that incorporate demand and supplier disruption information updates, while considering both risk-neutral and risk averse (CVaR) objectives. These models assist organizations with varying risk attitudes in achieving maximum performance by optimally selecting a procurement portfolio based on the availability and quality of updated information. Through analytical solutions and extensive numerical studies, our findings offer novel insights to organizations and policymakers, empowering them to enhance their supply chain resilience during critical supply disruption situations. Therefore, this research has broader implications and aligns with the multiple sustainability objectives outlined in the Sustainable Development Goals (SDGs).

**Keywords** Procurement management · Supply disruption · Option contracts · Spot markets · Value of information · Risk

## 1 Introduction

Economic globalization, product proliferation, and rapid technological progress have led to volatile customer demand and shorter product life cycles. Despite recent advancements in forecasting techniques, accurately estimating market conditions remains a challenge for many industries. Supply-side imperfections also arise due to various risk factors, including natural disasters, machine breakdowns, exchange rate volatility, terrorism, pandemics, and more. Complex supply networks, coupled with industrial automation and decentralized manufacturing increase the vulnerability of supply chains to disruptions (Rujeerapaiboon et al.,

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2023). The resulting mismatches between supply and demand can have devastating outcomes for organizations and customers. For example, the Covid-19 pandemic and the global lockdown led to a combination of demand surges and supply disruptions revealing the fragility of many supply chains (Mohammad et al., 2022). Semiconductor chip shortages have severely impacted major companies in various industries such as Apple, Sony, Tesla, and BMW (Bailey, 2020; Mohammad et al., 2022; Ramani et al., 2022; Voas et al., 2021). The production disruptions, backlogs, and delays caused by the shortage have resulted in estimated global annual production losses of \$110 billion as of May 2021 (Wayland, 2021). These events reveal the need for organizations to implement state-of-the-art methodologies to effectively balance customer demand and supply while minimizing risks.

In recent decades, the fields of risk management have garnered noteworthy interest from researchers and practitioners, supporting global efforts towards building a more resilient supply chain ecosystem (Wu, 2016). These efforts led to supply chain risk assessment and mitigation strategies to help companies streamline their operations (Choudhary et al., 2023). One such strategy is diversification flexibility, which suggests having multiple sources of supply to mitigate the risk of disruptions. In this study, we focus on two procurement modes: the forward contract and the option contract. The adoption of option contracts has been observed in various industries, including apparel, electronics, and fresh produce (Barnes-Schuster et al., 2002; Eppen & Iyer, 1997; Nagali et al., 2008). The forward contract is less expensive but has an extended procurement lead time and is subject to disruptions. The option contract is more expensive but offers greater reliability with a much shorter lead time. In our study, option contracts provide retailers with the opportunity to share the risks associated with supply disruptions, demand fluctuations, and price volatility with the suppliers.

In combination with forward and option contracts, firms in various industries also seek additional supplies from other resources. Spot markets have been actively employed as an instrument to hedge supply–demand inequity risks in the supply chains. While spot markets offer products with negligible lead time, they come with higher and volatile prices to compensate the extra flexibility they offer. The joint usage of spot markets and capacity options promotes the efficient risk management for commodities such as metal and agricultural products. In addition to commodities, the use of option contracts together with the spot market is also observed in various industries such as electronics and energy (Hou & Zhao, 2012; Kim, 2005). For example, Hewlett Packard (HP) invests in 50% forward contracts, 35% option contracts, and leaves 15% to spot market purchases (Billington et al., 2002). Moreover, fast-track suppliers, who may secure production for its customers in extremely short lead times, may also serve as a spot market for various industries. Zhao et al. (2018) presents the case of a trading company, SADL, from China offering design, merchandising, manufacturing, and delivery service to apparel retail brands in Europe. SADL's competitive strategy is to provide production a very short lead time (e.g., two to three weeks) and a very small quantity (e.g., 2–3 pieces). In return, this company receives a relatively high profit margin and quick credit payment.

In this study, we allow the supplier's disruption status to influence the spot market price distribution. In practice, when the supplier is the primary provider of a specific part's global supply; disruptions at that supplier can substantially impact the spot market price and introduce volatility in the market. Even if the supplier is not a key player, the supplier's disruption may be the indicator of a larger scale supply chain catastrophe signaling a joint supply and demand shock. For example, recently the Covid-19 pandemic has contributed enormously to market price instabilities in several industries. Semiconductor shortages led to price increases in the bill of materials, which were then passed on to consumer products such as TVs or PCs (Kaur, 2021; Knight, 2023; Mohammad et al., 2022). According to the Food Security Portal

(FSP), who monitors the effects of the global pandemic on food prices, certain food commodities have experienced excessive price variability due to the pandemic (e.g., cotton's spot price increased by 23% between March and July 2020) (FSP, 2020). Similarly, the spot market prices of the most metal commodities felt the negative shock. To illustrate, Cameco, one of the largest global providers of uranium, announced that pandemic disrupted their operations, leading to a more than 35% increase in spot price between March 2020 and July 2020 (Cameco, 2020). Analyzing spot market prices in conjunction with the supply disruptions is therefore critical for firms to optimize their operations and to achieve maximum performance.

Practitioners are often hesitant to fully adopt risk mitigation strategies such as excessive capacity due to the need for expensive investments. The solution to avoid the inefficiency is to invest in flexibility, rather than redundancy in the supply chains. Organizations can exploit the benefits of the embedded flexibility in the system by closely monitoring supply or demand patterns and employing data in smart ways. Effective utilization of advance demand information enhances the efficiency and resistance of production/inventory systems. Similarly, firms can employ various strategies to obtain more information about the supplier's disruption status before the scheduled shipment date. For example, the buyer may continuously track shipments from the supplier to monitor if the earlier shipments are received on time and if they are complete. This information enables the buyer to update the firm's assessment of the disruption probability and make informed decisions. In our study, option contracts serve as a reactive mechanism in response to updated information. Despite being more expensive, the retailer benefits from the ordering flexibility provided by the reserved capacity, which can be exercised in response to updated information about the supplier's disruption status, demand conditions, and spot market prices.

The supply chain procurement literature traditionally considers risk-neutral operating conditions and use expected profit measure to assess the overall value of the procurement portfolio. On the other hand, maximizing the expected profit in the long run may not always be desirable from the practical perspective especially when demand and spot market price are highly unpredictable and supply disruption risks are present. In such situations, minimizing a downside risk measure could be more appropriate. To address this, our analysis incorporates Conditional Value at Risk (CVaR), a popular downside risk measure. By integrating CVaR into our models, we can capture the differences in procurement strategies between risk averse and risk-neutral decision makers, considering varying levels of information updates.

In 2015, the United Nations presented a global agenda to tackle economic, social, and environmental challenges and to achieve sustainable development by 2030 (United Nations, 2015). For this purpose, the organization formulated clear goals via 17 SDGs. Our study aims to contribute to this broad sustainability program by aligning with multiple SDGs. The Covid-19 pandemic highlighted the vulnerability of the supply chains due to extended supply shortages and unprecedented demand surges. By utilizing multi-stage, risk averse stochastic programming models that incorporate real-time updates on customer demand and supplier disruptions, we propose novel decision-making tools to enhance the resilience of supply chains in critical times. Therefore, our efforts align well with SDG 9 (Industry, Innovation, and Infrastructure). We optimize procurement strategy of the firms via flexible option contracts modeling the interrelations between disruptions and market dynamics. Considering economic and environmental consequences, our models result lower excesses and shortages promoting responsible consumption and procurement, aligning with SDG 12 (Responsible Consumption and Production). Additionally, during humanitarian crises like natural disasters, efficient procurement and allocation of critical items become critical to alleviate human suffering and cut inequalities. Therefore, when employed within humanitarian supply chain contexts, our research supports SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

The rest of the paper is organized as follows. Section 2 provides a comprehensive review of related literature for this problem. Section 3 introduces the notation, assumptions, and presents the mathematical models for both risk averse and risk-neutral decision-makers. Section 4 introduces special cases for the problem and provides analytical solutions. The results of numerical tests are presented in Sect. 5, and finally, Sect. 6 concludes the study.

## 2 Literature review

The literature review for this study can be classified into four streams. The first stream provides an overview of the literature on supply chain disruptions and explores strategies to enhance resilience. The second stream focuses on studies that consider multiple procurement channels, such as multiple or unreliable suppliers, option contracts, or spot markets. The third stream of literature examines the value of demand and supply information in replenishment decisions. Finally, the fourth stream overviews the literature on sustainability efforts in supply chain management in connection with the SDGs.

Several studies have investigated the impacts of disruptions and risk mitigation strategies for supply chains. Hendricks and Singhal (2005) empirically study the effect of supply chain disruptions on long-run stock price performance and equity risk of the firm. Drawing on the literature and the case studies from the industry, Sodhi and Tang (2012) summarize modelling-based approaches to mitigate supply chain risk. Snyder et al. (2016) provide an extensive review of OR/MS models for supply chain disruptions. More recently, Katsaliaki et al. (2022) conduct a review on supply chain disruptions and risk mitigating strategies. According to Lee (2004), supply chains that prioritize efficiency often deteriorate over time, while companies that focus on agility, adaptability, and alignment gain a competitive edge. Lee (2004) emphasizes the importance of flexibility in building resilient and responsive supply chains. However, determining the optimal level of flexibility to mitigate risks remains challenging due to insufficient data and elaborate cost–benefit analysis. Tang and Tomlin (2008) address this problem by presenting a unified framework and stylized models that demonstrate the strategic value of implementing a risk reduction program with a relatively low level of flexibility. Simchi-Levi et al. (2018) investigate a hybrid strategy that combines process flexibility and finished goods inventory to mitigate supply chain risks. They propose a two-stage robust optimization model that considers inventory allocation before disruptions and production quantities determination after disruptions. Mehmanchi et al. (2021) study the worst-case performance of flexibility designs under demand and supply uncertainties, including disruptions. They introduce the Plant Cover Index under Disruptions (PCID) as a measure for assessing required plant capacity after disruptions. More recently, Rujerapaboon et al. (2023) investigate the resilience of long chains when there is a positive probability of plant disruptions, proposing a simple greedy algorithm to describe the performance of the chain under different disruption and demand scenarios.

The second stream of literature focuses on studies that investigate the use of multiple procurement channels, including multiple/unreliable suppliers, option contracts, or spot markets. Seifert et al. (2004) examine a single-period replenishment model and determine the optimal order quantity for purchasing through option contracts and spot markets. Kleindorfer and Wu (2005) study multiple unreliable suppliers through option contracts with a random spot market price, assuming price and quantity dependent demand functions. Dada et al. (2007) consider a firm with multiple reliable and unreliable suppliers, proposing the "cost first reliability second" strategy for risk-neutral supplier selection. We show that this insight

is only valid for risk-neutral models without correlation between spot market price and supplier disruption status. Martínez-de-Albeniz and Simchi-Levi (2009) analyze a single-period portfolio problem, investigating suppliers' competitive behavior and assuming constant spot market prices where the spot market acts as a backup supplier with a large execution price but zero reservation price. Fu et al. (2010) study a similar single-period model, where after the realization of random demand and spot price, the firm has recourse options to utilize either option contracts or the spot market. Merzifonluoglu (2015a, 2015b, 2017) and Mohammadi-vojdani et al. (2022) study diverse versions of single-period procurement models involving forward contracts, option contracts, and spot markets, developing optimal properties for supplier selection and offering insights. Luo and Chen (2017) examine the impact of option contracts in a random yield supply chain with a spot market. They show that option contracts can coordinate the supply chain, but they also identify conditions where the benefits of option contracts may not be significant. Wang et al. (2019) study the effects of customer returns and bidirectional option contracts on a newsvendor firm's refund price and order decisions. Xue et al. (2020) analyze the use of option contracts and order commitment contracts to mitigate supply disruption risks in decentralized supply chains. Fan et al. (2020) consider the application of an option contract in a buyer-led supply chain where both the buyer and supplier are risk averse, using value-at-risk (CVaR) optimization. Luo et al. (2021) develop a game-theoretic model to assess the value of bidirectional options in a supply chain. While most of the literature cited in this stream focuses on two-stage models where uncertainty in supply, demand, and spot market prices are resolved upon exercising options, our study presents three-stage stochastic programming models where the buyer receives imperfect information on demand and supply uncertainty in the second stage and reacts using option contracts.

Several studies have focused on inventory management with demand information updates or forecast revisions. We refer the readers to the detailed review and comments in the book by Sethi et al. (2005) and the references therein. This approach is developed and used by Gurnani and Tang (1999), Gallego and Ozer (2001), Sethi et al., (2001, 2003, 2004, 2005), Karaesmen et al. (2004) and others. Eppen and Iyer (1997) consider a backup agreement which is similar to an option contract where demand forecast in the second stage can be revised by the demand observed in the first stage and demonstrated that backup agreements have a substantial impact on the expected profit and the committed quantity. Barnes-Schuster et al. (2002) develop analogous models, and their results are in line with Eppen and Iyer (1997) in terms of the value of the demand information when using option contracts. Other research on the option-related contracts that allow order adjustments after updating demand information includes Brown and Lee (2003), Bassok and Anupindi (2008), Moynadeh and Nahmias (2000) and Huang et al. (2005). Guo et al. (2016) study a firm that procures a product from a regular supplier with supply disruption and yield risks and a backup supplier with capacity reservation option. They find out that the value of the reservation option depends on demand uncertainty and whether the reservation capacity is exercised before or after demand realization. Zhao et al. (2018) study spot markets in a two-stage model to explore the significance of supply option contracts with both demand information updating and a stochastic spot market. Our models differ from theirs in several aspects such as existence of an unreliable supplier, risk averse decision maker and supplier disruption information updates. We also explicitly model the correlation between the spot market price and the disruption status of the supplier explicitly. Furthermore, most of the supply contract literature with information updates considers risk neutral models except Buzacott et al. (2011) who develop a newsvendor model for investigating risks of commitment–option contracts with information updates using a mean–variance framework. In our study, we take a different

approach by employing CVaR as a risk measure and allowing the buyer to order from a supplier facing complete disruption, as well as from a spot market with volatile prices.

Additional literature exists focusing the value of information in the face of disruption risk. Considering a setting with a single supplier and a single buyer, Yang et al. (2009) examine the effects of information asymmetry and assess the value of disruption risk information. Yang et al. (2012) study a similar setting and suggest that the firm tends to have multiple sources of supply under better information. Saghaian and Van Oyen (2012) investigate the value of disruption information in a two-product setting, where there is a primary, unreliable supplier for each product and a reliable supplier that can produce both products through an options contract. Kole and Bakal (2017) examine the replenishment strategy of a buyer with two suppliers, a primary one and a reliable one. They study the value of supplier reliability information by considering full and partial information settings. Our study differs from the aforementioned studies in several aspects. Firstly, we aim to develop a comprehensive procurement portfolio consisting of a regular supplier subject to disruption, an option contract, and a volatile spot market. Moreover, we evaluate the value of information, specifically supply disruption and demand information, in a setting where the decision maker is risk averse. Furthermore, we incorporate a modeling approach that captures the relationship between supply disruption status and the uncertain spot market prices.

Lastly, we review studies that have contributed to the literature on sustainable supply chain management (SSCM) and their role achieving SDGs. Brandenburg et al. (2014) provide a systematic literature review to outline quantitative models in the SSCM research. López-Castro and Solano-Charis (2021) conduct a literature review from 2010 to 2021 emphasizing supply chain network design models with sustainability and resilience criteria. More recently, Agrawal et al. (2022) carry out a literature review that assessed the role of Sustainable Development Goals (SDGs) in different stages of the supply chain. They specifically investigate how SDGs are incorporated into product design, procurement, manufacturing, transportation, distribution, reverse logistics, and waste management. Moreover, Genc (2021) studies the implementation of the SDGs in supply chains, considering the influence of behavioral consumers. Rajesh (2021) and Chowdhury et al. (2021) address the challenges posed by global events such as the pandemic and offer suggestions to achieve sustainable supply chains. Khan et al. (2022) assess the risk of supply chain disruption due to Covid-19 with fuzzy VIKORSort. In response to severe supply disruptions due to Covid-19 pandemic, Nasir et al. (2022) explores the relations among various factors affecting supply chain viability in reaching SDGs. The recent literature highlights vulnerability of the modern supply chains and calls for sustainability and resilience. Complementing the recent literature; our study provides optimization models, solution approaches and actionable insights aligning SDGs (3), (9), (10) and (12).

To the best of our knowledge, this is the first study that simultaneously concentrate on the supply disruption, option contracts and volatile spot markets in the presence of information updates. This paper fills the research gap by examining value of information revisions on both demand forecasts and supplier disruption risks on portfolio selection decisions. The current study manages to characterize procurement strategy of a firm by providing optimality properties for the special case where the decision maker is risk neutral. These optimality properties also lead to novel managerial insights in portfolio selection decisions when spot market price is volatile and affected by disruption status of the supplier. Moreover, we have developed tractable scenario-based optimization formulations to either maximize expected profit or minimize risk (CVaR). We conducted an extensive numerical study that resulted in actionable insights regarding the value of information for decision-makers with varying

degrees of risk. Through numerical tests, we validated our analytical results, providing further evidence of the effectiveness and robustness of the proposed modeling and solution approach.

### 3 Preliminaries and the mathematical models

This study considers a single selling period model where the firm has two supply options namely a primary supplier and a backup supplier to satisfy random customer demand,  $D$ . The decision sequence is outlined in Fig. 1, and the notation used in the study is introduced in Table 1.

Before the selling season, the buyer must sign a forward contract with the primary supplier determine the order quantity ( $q_c$ ). The primary supplier is relatively less expensive (with the unit cost of  $c$ ) while it is subject to a risk of complete disruption with the probability of  $p$ . In our models,  $Y$  is a Bernoulli random variable denoting supply disruption for a primary supplier where  $Y = 1$  confirms disruption. If the primary supplier is disrupted, the firm does not receive any shipment from the primary supplier thus pays nothing.

In addition to the primary supplier, the buyer has the option to sign an option contract with the backup supplier, referred to as the "secondary supplier" or "reliable supplier." This option contract allows the buyer to reserve a certain capacity ( $k$ ) at a unit reservation cost of  $h$ . The reserved capacity can be later utilized by the buyer at a unit cost of  $e$ . It is assumed that the combined cost of the reservation and execution from the backup supplier, denoted as  $h + e$ , is higher than the unit cost from the primary supplier,  $c$ .

At the end of the selling season, any remaining demand that the buyer needs to satisfy is obtained from the spot market. The spot market is assumed to have infinite capacity, and its price is subject to volatility. The distribution of spot market prices depends on the disruption status of the supplier. Let  $g_Y(s)$  represent the density function and  $G_Y(s)$  represent the cumulative distribution function (cdf) of the spot market prices when the supplier's disruption status is  $Y$ . The mean of the spot market prices when the supplier is not disrupted is denoted as  $\mu_s^0$ , and the mean when the supplier is disrupted is denoted as  $\mu_s^1$ . Similarly, the variances are denoted as  $\sigma_s^0$  and  $\sigma_s^1$ , respectively. It is assumed that  $\mu_s^1 \geq \mu_s^0 > h + e$  and  $\sigma_s^1 \geq \sigma_s^0$ . This implies that the spot market price distribution has *two modes* depending on the supplier's disruption status. When the supplier is disrupted, mean and variance of spot market price are potentially higher.

Let  $t_1$ ,  $t_2$ , and  $t_3$  denote the epochs representing Stage 1, Stage 2, and Stage 3. At  $t_1$  the buyer purchases an order quantity from the primary supplier ( $q_c$ ) with the unit cost  $c$  and reserves some capacity at the backup supplier ( $k$ ) with the unit reservation cost of  $h$ . At  $t_2$ , the buyer receives signals  $U$  and  $I$  that *update* the demand distribution and the supplier disruption probability respectively. Based on the available information on supply disruption status and customer demand, the buyer orders  $q_e$  from the reserved capacity of the backup (or "executes options"), i.e.,  $q_e \leq k$ . In practice,  $t_2$  can be considered as the start of the selling season. At  $t_3$  uncertain customer demand ( $D$ ) and spot market price ( $S$ ) are fully realized. Any unsatisfied demand by the forward contract and the option contract is satisfied by the spot market at this stage ( $q_s$ ). The firm receives unit revenue  $r$  from each unit of satisfied demand  $D$ . In practice,  $t_3$  can be considered as the end of the selling season.

Note that signals received in the second stage are just an indication and could be imperfect. For the demand, at  $t_1$  the joint density  $\psi(d, u)$  is available. The signal  $U$  is observed at  $t_2$  with the marginal density  $\lambda(u)$ . If the observed value of  $U$  is  $u$ , then we obtain the updated demand density as the conditional density  $f(d|u) = \psi(d, u)/\lambda(u)$  for  $\lambda(u) > 0$ . We let the

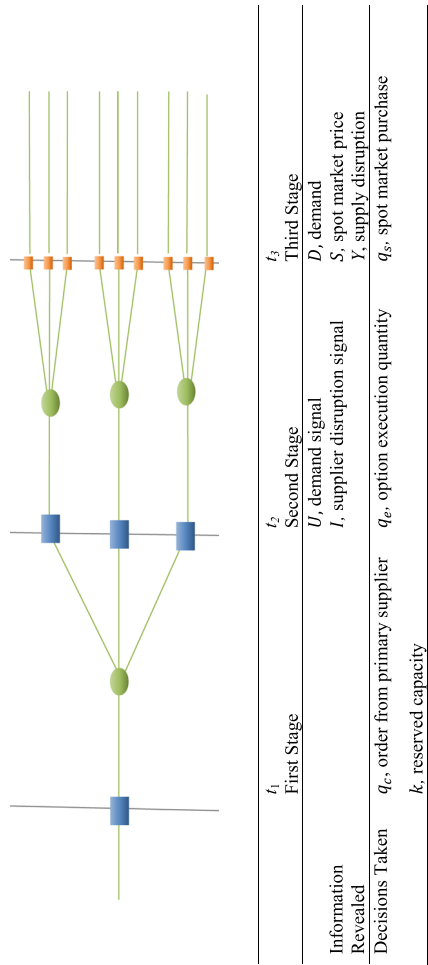


Fig. 1 Decision sequence and information updates in the three stage stochastic programming model

**Table 1** Nomenclature*Decision variables*

|       |                                             |
|-------|---------------------------------------------|
| $q_c$ | Order quantity from the primary supplier    |
| $k$   | Capacity reserved at the backup supplier    |
| $q_e$ | Quantity ordered from the reserved capacity |
| $q_s$ | Quantity ordered from the spot market       |

*Problem parameters*

|     |                                                        |
|-----|--------------------------------------------------------|
| $c$ | Unit ordering cost from the primary supplier           |
| $e$ | Cost of ordering per unit from the secondary supplier  |
| $r$ | Revenue per unit sold                                  |
| $h$ | Cost of reserving per unit from the secondary supplier |
| $p$ | Probability of supplier disruption                     |

*Random variables*

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| $Y$ | Bernoulli variable denoting supply disruption for a primary supplier; $Y = 1$ shows disruption |
| $D$ | Customer demand                                                                                |
| $S$ | Spot market price                                                                              |
| $U$ | The signal that updates the demand distribution                                                |
| $I$ | The signal that updates the supplier disruption probability                                    |

cumulative distributions corresponding to the densities  $f(d|u)$ ,  $\psi(d, u)$  and  $\lambda(u)$  are denoted by  $F(d|u)$ ,  $\Psi(d, u)$  and  $\Lambda(u)$  respectively. When the information update is worthless, then  $f(d)$  and cdf  $F(d)$  denote the demand distribution as observed at  $t_1$  and  $t_2$ .

A signal on the supply disruption ( $I$ ) is also observed at  $t_2$ . We assume that the firm has an initial (unbiased) prediction on the disruption probability ( $p$ ). We assume that the *signal* can return values of 0 or 1 implying cases of no-disruption and disruption. We assume that this *signal* is imperfect and only  $100\gamma\%$  accurate. This implies that when the supply will be disrupted in stage 3 ( $Y = 1$ ), the test will return  $I = 1$  with probability of  $\gamma$ . We then let  $p(i)$  denote the conditional probability of supplier disruption after the evidence  $I$  is taken into account. We employ Bayes' theorem to determine the posterior probabilities,  $p(1)$  and  $p(0)$ :

$$\begin{aligned}
 p(1) &= P(Y = 1|I = 1) = \frac{P(I = 1|Y = 1)P(Y = 1)}{P(I = 1|Y = 1)P(Y = 1) + P(I = 1|Y = 0)P(Y = 0)} \\
 &= \frac{\gamma p}{\gamma p + (1 - \gamma)(1 - p)} \\
 p(0) &= P(Y = 1|I = 0) = \frac{P(I = 0|Y = 1)P(Y = 1)}{P(I = 0|Y = 1)P(Y = 1) + P(I = 0|Y = 0)P(Y = 0)} \\
 &= \frac{(1 - \gamma)p}{(1 - \gamma)p + \gamma(1 - p)}
 \end{aligned}$$

When the signal perfectly describes the supplier's disruption status, then  $\gamma = 1$ . This means that when the test indicates disruption ( $I = 1$ ), the probability of disruption equals to unity,  $p(1) = 1$ . Similarly, when the test indicates no-disruption ( $I = 0$ ), the probability of disruption equals to zero,  $p(0) = 0$ . When the test is worthless ( $\gamma = 0.5$ ), then  $p(0) = p(1) = p$ . This suggests that the "posterior probability" is same as the firm's initial prediction for the supplier's disruption probability.

We next introduce two fundamental models for this problem with risk neutral and risk averse objectives.

### 3.1 Risk neutral approach

We next present our model maximizing expected profit which we refer to as Model I.

#### 3.1.1 Model I

$$\pi^* = \underset{q_c, k \geq 0}{\text{Maximize}} \left\{ -kh + E \left( \underset{0 \leq q_e \leq k}{\text{Maximize}} \Pi(q_e; k, q_c, I, U) \right) \right\} \text{ where } \Pi(q_e; k, q_c, I, U) = rE[D|U] - eq_e - (p(i)\mu_s^1 E[(D - q_e)^+ | U] + (1 - p(i))\mu_s^0 E[(D - (q_c + q_e))^+ | U]).$$

This is a three-stage stochastic programming model. Our core motivation is to find the optimal order quantity from the primary supplier ( $q_c$ ) and the optimal reservation quantity at the backup supplier ( $k$ ).

We start our analysis with the second stage problem first. At this point, the firm has full knowledge of the first stage variables ( $q_c$  and  $k$ ). In addition, the demand market signal ( $U$ ) and the supply disruption signal ( $I$ ) are already revealed. The decision maker has the updated demand distribution  $F(d|u)$  and the updated supplier disruption probability ( $p(i)$ ) at this point. We let  $\mu_S(i) = p(i)\mu_s^1 + (1 - p(i))\mu_s^0$  denote the expected spot market price implied by the disruption signal  $i$ . The decision maker needs to determine the option execution quantity  $q_e$  by solving an extension of the newsvendor problem with maximum ordering limitation ( $k$ ). The Stage 2 problem then becomes the following.

$$\min_{0 \leq q_e \leq k} eq_e + p(i)\mu_s^1 \Delta_u(q_e) + (1 - p(i))\mu_s^0 \Delta_u(q_c + q_e)$$

In the formulation above, we let  $\Delta_u(x)$  denote a loss function which shows the expected shortfall in Stage 3 given demand signal  $u$ , i.e.,  $\Delta_u(x) = E[(D - x)^+ | u] = \int_x^\infty (d - x)F(d|u)$ . It is then straightforward to show that the optimal execution quantity would be either 0,  $\Theta$  or  $k$  based on the following condition:

$$\begin{aligned} q_c \leq q' &\Rightarrow q_e = k \\ \text{If } q' < q_c \leq q'' &\Rightarrow q_e = \Theta. \\ q'' < q_c &\Rightarrow q_e = 0 \end{aligned}$$

Given  $q_c$  and  $k$ , the values of  $\Theta$ ,  $q'$  and  $q''$  can be determined by solving Eqs. (1), (2) and (3) respectively.

$$\frac{\mu_s^1 p(i)}{\mu_S(i)} F(\Theta) \Big|_u + \frac{\mu_s^0 (1 - p(i))}{\mu_S(i)} F(q_c + \Theta) \Big|_u = \frac{\mu_S(i) - e}{\mu_S(i)} \quad (1)$$

$$\frac{\mu_s^1 p(i)}{\mu_S(i)} F(k) \Big|_u + \frac{\mu_s^0 (1 - p(i))}{\mu_S(i)} F(q' + k) \Big|_u = \frac{\mu_S(i) - e}{\mu_S(i)} \quad (2)$$

$$\frac{\mu_s^1 p(i)}{\mu_S(i)} F(0) \Big|_u + \frac{\mu_s^0 (1 - p(i))}{\mu_S(i)} F(q'') \Big|_u = \frac{\mu_S(i) - e}{\mu_S(i)} \quad (3)$$

Although simultaneous treatment of market and supply information to adjust the level of flexible ordering makes sense from the practical point of view, resulting models are unavoidably complex. We show that the first stage variables and information signals, the decision maker can easily solve the second stage problem and compute  $q_e$ . However, it is still challenging to determine the optimality conditions for the first stage variables. Therefore, in Sect. 4,

we study optimization models designed for certain information settings only (perfect and worthless information).

To tackle the inherent complexity of the models, we next provide optimization models for the general setting assuming that the future can be described by a finite number of scenarios in the second and the third stage. With this assumption, the *three-stage* stochastic linear programs can be modelled as large linear programming problems (deterministic equivalent linear programming model). This formulation would particularly be useful to incorporate risk aversion into our models as it is demonstrated in Sect. 3.2.

We let  $W$  denote the number of possible scenarios in the second stage. Note that scenario  $w$  in the second stage clearly describes the supply disruption signal ( $I_w$ ) and the demand signal ( $U_w$ ). We let  $P_w$  denote the probability of occurrence of scenario  $w$  in the second stage. We let  $T$  denote number of possible third stage scenarios for each second stage scenario  $w$ . The pair  $(w, t)$  clearly determines the supplier's disruption status ( $Y_{wt}$ ), spot market price ( $S_{wt}$ ) and customer demand ( $D_{wt}$ ) in the third stage. We let  $P_{wt}$  denote the probability of occurrence of scenario  $w$  in the second stage and scenario  $t$  in the third stage.

We next present Model I-s which transforms Model I to a scenario-based representation.

### 3.1.2 Model I-s

$$\text{Maximize } \sum_{w=1}^W \sum_{t=1}^T P_{wt} \pi_{wt} \quad (4)$$

Subject to

$$\pi_{wt} = rD_{wt} - (kh + cq_c(1 - Y_{wt}) + e(q_e)_w + S_{wt}(q_s)_{wt}), \quad t = 1, \dots, T; w = 1, \dots, W \quad (5)$$

$$q_c(1 - Y_{wt}) + (q_e)_w + (q_s)_{wt} \geq D_{wt}, \quad t = 1, \dots, T; w = 1, \dots, W \quad (6)$$

$$(q_e)_w \leq k, \quad w = 1, \dots, W, \quad (7)$$

$$k, q_c, (q_e)_w, (q_s)_{wt} \geq 0. \quad t = 1, \dots, T; w = 1, \dots, W. \quad (8)$$

The objective function (4) maximizes the overall expected profit, the constraint set (5) defines the profit for the third stage scenario  $(w, t)$ , i.e.,  $\pi_{wt}$ . The constraint set (6) and (7) determine the spot market purchase quantity in the third stage while the constraint set (8) addresses nonnegativity restrictions.

Note that, scenario-based formulations can effectively be utilized when the probability distributions of random variables are discrete, and all possible future states can be described by a finite number of scenarios. On the other hand, the actual number of possible scenarios in such a setting could be prohibitive. Furthermore, demand and spot market prices are generally described by continuous random distributions. Consequently, in the numerical study we describe utilization of Monte Carlo simulation to create future scenarios (for signals and random variables based on the signals). This is a well-known and effective technique and allows us work on various random number distributions while reducing the scenario set to a controllable size.

### 3.2 Risk averse approach

The decision models presented in Sect. 3.1 describe how the expected profit function of the firm is maximized when procurement decisions are under consideration. On the other hand, it is often undesirable to consider only the maximization of the expected profit when the decision maker also targets to avoid large financial losses. Our problem involves multiple, uncertain elements (demand, supply disruptions and spot market price), the variance of the resulting profit may turn out to be quite large. We incorporate risk in our problem employing the Conditional Value-at-Risk (CVaR) as a risk measure. For a given significance value,  $\alpha$ , the Value-at-Risk, VaR, is equal to the largest value making sure that the probability of obtaining a profit less than VaR is lower than  $(1 - \alpha)$ . This means, the VaR is the  $\alpha$ -quantile of the profit distribution. CVaR is derived by taking a weighted average between the VaR and losses exceeding the VaR. Although there are other risk approaches that can be applied to supply chain optimization problems, such as VaR and mean–variance, we chose CVaR as it is proven to be coherent and consistent with the second order stochastic dominance (see Rockafellar & Uryasev, 2000, 2002; Uryasev, 2000; Artzner et al., 1999; Pflug, 2000). When the risk is accounted for multistage stochastic programming such as the model under consideration; there is no natural or obvious way of modelling and measuring risk. The difficulty in extending risk measures to the multistage setting has been discussed by several studies (please see Homem-de-Mello & Pagnoncelli, 2016 and the references therein).

We propose the following decision model to maximize the CVaR value of the overall profit distribution.

#### 3.2.1 Model I-CVaR-s

$$\text{Maximize } \psi - \frac{1}{(1 - \alpha)} \sum_{w=1}^W \sum_{t=1}^T P_{wt} \xi_{wt} \quad (9)$$

Subject to (5)–(8)

$$\xi_{wt} \geq \psi - \pi_{wt}, \quad t = 1, \dots, T; w = 1, \dots, W, \quad (10)$$

$$\xi_{wt} \geq 0, \quad t = 1, \dots, T; w = 1, \dots, W. \quad (11)$$

In the formulation above, the objective function (9) maximizes the CVaR of the third stage profit. Constraint set (10) defines as the tail loss for scenario  $(w, t)$ , which is the amount by which the profit in scenario  $(w, t)$  is short of  $\psi$  (representative of VaR). Constraint set (11) makes sure that  $\xi_{wt}$  is a continuous nonnegative decision variable.

In the numerical study, we solve Model I-CVaR-s for varying  $\alpha$  values and present insights on how risk aversion affects the behavior of the decision maker towards new information when procurement decisions are made. We next concentrate our efforts on the risk neutral decision maker and characterize the optimal procurement quantities associated with certain extreme cases.

### 4 Special case: risk neutral decision maker

In this section, we consider a risk neutral decision maker who maximizes the expected profit of the firm when taking procurement related decisions. We will study the value of

**Table 2** Problem variants for the special case

| Model Name                                    | Stage when information revealed |           |
|-----------------------------------------------|---------------------------------|-----------|
|                                               | Stage 2                         | Stage 3   |
| Perfect Information (PI)                      | $Y, D$                          | $S$       |
| Perfect Supplier Disruption Information (PSI) | $Y$                             | $D, S$    |
| Worthless Information (WI)                    | –                               | $Y, D, S$ |

demand and supply disruption information in further detail for extreme cases; providing mathematical formulations, optimality conditions and managerial insights (see Table 2 for problem variants).

#### 4.1 Perfect information

In this subsection, we study an extreme case for which both the demand and disruption information revision is perfect. The perfect information revision represents a setting such that the demand  $D$  and supplier reliability  $Y$  can be completely determined by the information observed at Stage 2. Note that spot market price is still revealed in Stage 3. We refer to this model as “Perfect Information”. Our optimization problem can be written as

$$\pi^* = \max_{q_c, k \geq 0} \{kh + pE[\Pi(q_e; k, q_c, 1, D)] + (1 - p)E[\Pi(q_e; k, q_c, 0, D)]\}.$$

In the formulation,  $\Pi(q_e; k, q_c, Y, D)$  denotes the expected profit of the second stage scenario which we will derive next for the disruption ( $Y = 1$ ) and no-disruption ( $Y = 0$ ) cases.

##### 4.1.1 Disruption

In case of a disruption, the buyer does not receive any product from the primary supplier. This case occurs with the probability of  $p$ . The buyer must use either spot market or the capacity reserved at the secondary supplier to fulfil the demand. Since the execution cost of the option contract is less than or equal to expected spot market price ( $e < \mu_s^1$ ), the firm would first utilize the reserved capacity to satisfy demand. The remaining demand would be satisfied by the spot market. We first let  $\Delta(x)$  denote the loss function denoting the expected shortfall in Stage 3, i.e.,  $\Delta(x) = E[(D - x)^+] = \int_x^\infty (d - x)F(d)$ . We can then state the expected execution quantity from the backup as

$$E[q_e] = \int_{-\infty}^k dF(d) + \int_k^\infty kF(d) = \mu_d - \Delta(k).$$

Similarly, expected spot market purchase becomes the following

$$E[q_s] = \int_k^\infty (d - k)F(d) = \Delta(k).$$

Expected profit function for the disruption case is then written as follows:

$$E[\Pi(q_e; k, q_c, 1, D)] = r\mu_D - (e\mu_d + (\mu_s^1 - e)\Delta(k)). \quad (12)$$

### 4.1.2 No-disruption

In this setting the buyer receives  $q_c$  from the primary supplier and will satisfy the rest of demand from the spot market or the reserved capacity. In this case, expected spot market price is  $\mu_s^0$ . Since  $\mu_s^0 > e$ , the buyer again first uses the reserved capacity. Expected execution quantity from the reserved capacity can be expressed as:

$$E[q_e] = \int_{q_c}^{q_c+k} dF(d) + \int_{q_c+k}^{\infty} kF(d) = \Delta(q_c) - \Delta(q_c + k).$$

Expected spot market purchase in this setting can be calculated as:

$$E[q_s] = \int_{q_c+k}^{\infty} (d - (q_c + k))F(d) = \Delta(q_c + k).$$

Expected profit for this setting can then be written as:

$$E[\Pi(q_e; k, q_c, 0, D)] = r\mu_D - (cq_c + e\Delta(q_c) + (\mu_s^0 - e)\Delta(q_c + k)) \quad (13)$$

Combining (12) and (13), we can write the expected profit for this problem as following:

$$\pi = r\mu_D - kh - p[e\mu_d + (\mu_s^1 - e)\Delta(k)] - (1 - p)[cq_c + e\Delta(q_c) + (\mu_s^0 - e)\Delta(q_c + k)].$$

### 4.1.3 Properties of the optimal solution

In this section, the optimal order quantities from the primary supplier and the backup supplier are analyzed. The objective function is concave and continuously differentiable for this problem for both first stage variables. KKT optimality conditions are necessary and sufficient. The reader may see the Appendix for proofs. We next define the partial derivatives of the objective function with respect to first stage variables, i.e.,

$$\begin{aligned} \frac{\partial \pi}{\partial q_c} &= (1 - p)(-c + e(1 - F(q_c)) + (\mu_s^0 - e)(1 - F(q_c + k))), \\ \frac{\partial \pi}{\partial k} &= -h + (p(\mu_s^1 - e)(1 - F(k)) + (1 - p)(\mu_s^0 - e)(1 - F(q_c + k))). \end{aligned}$$

**Proposition 1** In case of perfect information, the optimal order quantity and optimal reservation capacity are given by.

- If  $p \leq \frac{\mu_s^0 \left( \frac{h}{\mu_s^0 - e} \right) - c}{\mu_s^0 \left( \frac{\mu_s^1 - e}{\mu_s^0 - e} \right) - c}$ , then  $q_c = F^{-1} \left( 1 - \frac{c}{\mu_s^0} \right)$  and  $k = 0$ .
- If  $p > \frac{(\mu_s^0 - e)(h + e - c)}{(\mu_s^1 - \mu_s^0)(c - e)}$  and  $c > e$ , then  $k = F^{-1} \left( 1 - \frac{h}{(p\mu_s^1 + (1 - p)\mu_s^0) - e} \right)$  and  $q_c = 0$ .
- Otherwise,  $(k, q_c)$  is a unique optimal solution given by the solution of  $\frac{\partial \pi}{\partial q_c} = 0$  and  $\frac{\partial \pi}{\partial k} = 0$ :  $\frac{e}{\mu_s^0} F(q_c) + \frac{\mu_s^0 - e}{\mu_s^0} F(q_c + k) = 1 - \frac{c}{\mu_s^0}$ ,  $\frac{p\mu_s^1}{p\mu_s^1 + (1 - p)\mu_s^0} F(k) + \frac{(1 - p)\mu_s^0}{p\mu_s^1 + (1 - p)\mu_s^0} F(q_c + k) = 1 - \frac{h}{(p\mu_s^1 + (1 - p)\mu_s^0) - e}$ .

Based on KKT conditions, Proposition 1 provides optimal ordering quantities for both suppliers. This proposition also characterizes a *critical threshold* that determines whether the reservation capacity from the backup supplier option should be utilized or not. This threshold

is given by  $p \leq \frac{\mu_s^0 \left( \frac{h}{\mu_s^0 - e} \right) - c}{\mu_s^0 \left( \frac{\mu_s^1 - e}{\mu_s^0 - e} \right) - c}$ . If the disruption probability is below this threshold, then it is

optimal not to reserve any capacity at the backup supplier. However, if the unit cost for the primary supplier is higher than the execution cost of the option contract and the disruption probability of the supplier exceeds the threshold value of  $\frac{(\mu_s^0 - e)(h + e - c)}{(\mu_s^1 - \mu_s^0)(c - e)}$ , the buyer then prefers to procure solely from the backup supplier. The reader should note that the threshold values presented by this proposition not only depends on the cost parameters of the suppliers but also on the spot market price distribution.

**Corollary 1** In case of perfect information, when the expected spot market price is correlated with the supplier's disruption status, the critical ratio of the backup supplier,  $CR_k = 1 - \frac{h}{(p\mu_s^1 + (1-p)\mu_s^0) - e}$ , increases by the supplier's disruption risk.

We can define  $CR_{q_c} = 1 - \frac{c}{\mu_s^0}$  and  $CR_k = 1 - \frac{h}{(p\mu_s^1 + (1-p)\mu_s^0) - e}$  as the newsvendor critical ratio for the primary and the backup supplier respectively. This ratio can be used to determine the order (or reservation) quantities when these suppliers are the single providers of the supply. Interestingly, the critical ratio of the backup supplier depends on the reliability of the primary supplier. In our analysis, we let  $\mu_s^0$  denote the mean spot market price when the supplier is not disrupted and  $\mu_s^1 = \mu_s^0 + \varepsilon$  denote the mean spot market price when the supplier is disrupted. When  $\varepsilon > 0$ , we assume that the supply disruption leads to a higher expected spot market price. We can then formulate the critical ratio for the backup as  $CR_k = 1 - \frac{h}{\mu_s^0 + p\varepsilon - e}$ . Additionally, we define  $\mu_s$  as the expected spot market price where  $\mu_s = p\mu_s^1 + (1-p)\mu_s^0$ . Note that when the spot market distribution is not correlated with supplier's disruption status ( $\varepsilon = 0$ ), then the critical ratio becomes  $CR_k(0) = 1 - \frac{h}{\mu_s - e}$ . Therefore, as the correlation between spot market price and the disruption increases, the critical ratio, so the attractiveness of the backup supplier increases. When the primary supplier is extremely risky with higher  $p$  values, this effect is further amplified. Note that, when there is no correlation, both critical ratios ( $CR_k$  and  $CR_{q_c}$ ) are independent from the supplier's disruption risk. This result aligns with the literature's statement by Data et al. (2007) that "Cost is the order qualifier, and the reliability is the order winner." However, our analysis demonstrates that this result cannot be generalized when the spot market price is not independent of the reliability of the supplier.

**Corollary 2** In case of perfect information, when spot market price is independent from the supplier's disruption status, then the optimal order quantity and optimal reservation capacity are given by

- If  $p \leq \frac{\mu_s \left( \frac{h}{\mu_s - e} \right) - c}{\mu_s - c}$ , then  $q_c = F^{-1} \left( 1 - \frac{c}{\mu_s} \right)$  and  $k = 0$ .
- Otherwise,  $(k, q_c)$  is a unique optimal solution given by the solution of  $\frac{\partial \pi}{\partial q_c} = 0$  and  $\frac{\partial \pi}{\partial k} = 0$ , i.e.,  $\frac{e}{\mu_s} F(q_c) + \frac{\mu_s - e}{\mu_s} F(q_c + k) = 1 - \frac{c}{\mu_s}$ ,  $pF(k) + (1-p)F(q_c + k) = 1 - \frac{h}{\mu_s - e}$ .

Corollary 2 transforms Proposition 1 for the setting where the spot market distribution and the supplier disruption are not correlated ( $\varepsilon = 0$ ). Note that, due to the condition ( $e + h > c$ ) the buyer never orders exclusively from the backup supplier, regardless of the disruption risk as opposed to the setting where  $\varepsilon > 0$ . This result emphasizes the major impact of the correlation between supply disruption and spot market prices on the buyer's procurement decisions.

## 4.2 Perfect supplier disruption information

In this section, we study a setting where the decision maker has a certain information whether the supplier will be disrupted or not while the demand information revision is worthless. Therefore, demand forecast update has no improvement on the firm's ability to predict the demand.

Given the disruption status of the supplier ( $y = 0$  or  $y = 1$ ), the problem in Stage 2 is a classical newsvendor problem with order quantity  $q_e$  and possible initial inventory  $q_c(1 - y)$ . The newsvendor problem becomes  $\min_{0 \leq q_e \leq k} eq_e + \mu_s^y \Delta(q_c(1 - y) + q_e)$ . We next investigate the critical ratio for this newsvendor problem. Let  $\theta_y$  be the order quantity associated with the critical ratio, i.e.,  $\theta_y = F^{-1}\left(1 - \frac{e}{\mu_s^y}\right)$ .

If a supplier disruption occurs, the optimal number of units obtained from the option contract is  $q_e = \min\{\theta_1, k\}$ , as  $k$  is an upper limit on the order quantity at this stage. If there is no disruption, the optimal number of executed units is  $q_e = \max\{0, \min(\theta_0 - q_c, k)\}$ . Note that  $q_e \leq \theta_0 \leq \theta_1$ , therefore at most  $\theta_1$  units are executed in considering both cases. Therefore, it is not optimal to set  $k > \theta_1$ .

**Proposition 2** In case of perfect disruption information and worthless demand information, the reservation quantity at the backup,  $k$ , cannot exceed the quantity,  $\theta_1 = F^{-1}\left(1 - \frac{e}{\mu_s^1}\right)$ .

According to Proposition 2, when the supplier is disrupted, all capacity reserved at the backup is executed ( $q_e = k$ ) and the remaining demand will be satisfied with the spot market. If a disruption does not occur, the buyer receives  $q_c$  from the primary supplier and will satisfy the rest of its demand with spot market or the reliable supplier. We can analyze the profit function considering three possible solution settings in the optimal solution:

$$\pi = r\mu_D - kh - p(ek + \mu_s^1 \Delta(k)) - (1 - p) \left( cq_c + \begin{cases} \mu_s^0 \Delta(q_c), & \theta_0 < q_c \\ e(\theta_0 - q_c) + \mu_s^0 \Delta(\theta_0), & q_c \leq \theta_0 < q_c + k \\ ek + \mu_s^0 \Delta(q_c + k), & \theta_0 \geq q_c + k \end{cases} \right).$$

We next study the properties of the optimal solution for this problem. We first provide the partial derivatives of total expected profit with respect to  $q_c$  and  $k$ , respectively.

$$\frac{\partial \pi}{\partial q_c} = (1 - p) \begin{cases} \mu_s^0(1 - F(q_c)) - c, & \theta_0 < q_c \\ e - c, & q_c \leq \theta_0 < q_c + k \\ \mu_s^0(1 - F(q_c + k)) - c, & \theta_0 \geq q_c + k \end{cases}$$

$$\frac{\partial \pi}{\partial k} = -h + p(-e + \mu_s^1(1 - F(k))) + \begin{cases} 0 & \theta_0 < q_c \\ 0 & q_c \leq \theta_0 < q_c + k \\ (1 - p)(\mu_s^0(1 - F(q_c + k)) - e) & \theta_0 \geq q_c + k \end{cases}$$

Note that  $\pi$  is a continuously differentiable and concave function, and the KKT conditions are therefore necessary and sufficient for this setting. Proposition 3 presents possible solution strategies for this problem.

**Proposition 3** In case of perfect disruption information and worthless demand information, the optimal order quantity and optimal reservation capacity are given by

When  $e > c$

- If  $p \leq \frac{h}{\mu_s^1 - e}$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right)$ ,  $k = 0$ .
- If  $\frac{h}{\mu_s^1 - e} < p$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right)$ ,  $k = F^{-1}\left(1 - \frac{h+pe}{p\mu_s^1}\right)$ .
- When  $e \leq c$
- If  $p \leq \frac{h+e-c}{\mu_s^1 - c}$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right)$ ,  $k = 0$ .
- If  $\frac{h+e-c}{\mu_s^1 - c} < p \leq \frac{\mu_s^0(h+e-c)}{c(\mu_s^1 - \mu_s^0)}$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right) - k$ ,  $k = F^{-1}\left(1 - \frac{(h+e)-(1-p)c}{p\mu_s^1}\right)$ .
- If  $\frac{\mu_s^0(h+e-c)}{c(\mu_s^1 - \mu_s^0)} < p \leq \frac{h\mu_s^0}{e(\mu_s^1 - \mu_s^0)}$ ,  $q_c = 0$ ,  $k = F^{-1}\left(1 - \frac{h+e}{p\mu_s^1 + (1-p)\mu_s^0}\right)$ .
- If  $p > \frac{h\mu_s^0}{e(\mu_s^1 - \mu_s^0)}$ ,  $q_c = 0$ ,  $k = F^{-1}\left(1 - \frac{h+ep}{p\mu_s^1}\right)$ .

According to Proposition 3, if the execution cost of the option contract is higher than the regular unit ordering cost ( $e > c$ ) and the risk of the supplier disruption is lower than the threshold  $\left(\frac{h}{\mu_s^1 + e - c}\right)$ , the buyer has no incentive to reserve capacity at the reliable supplier. In this case, the primary supplier order quantity equals to the newsvendor solution. Note that this threshold decreases as the correlation between spot market price and disruption events increases. Therefore, when the spot market price is highly correlated with supplier disruptions, the decision maker considers the primary supplier as a single source only if the disruption risk is slim. When the expected marginal benefit of the option contract  $p(\mu_s^1 - e)$  is larger than its reservation cost  $h$ , some capacity is reserved at the reliable supplier.

When the execution cost of options is lower than the cost of buying from the primary supplier ( $e \leq c$ ), the threshold for reserving capacity at the backup for the risk of disruption is  $\left(\frac{h+e-c}{\mu_s^0 + \varepsilon - c}\right)$ . Note that this threshold is lower than the threshold for the setting ( $e > c$ ). This result is intuitive as option contract is more cost-efficient in this setting, so the buyer is more inclined to reserve capacity at the backup supplier. Similar to the previous setting, this threshold also decreases as the correlation of spot market price and disruption event increases.

The condition for utilizing the option contract depends on whether the marginal benefit of reserving optional capacity with the reliable supplier,  $p(\mu_s^1 - e) + (1-p)(c - e)$ , is greater than its reservation cost  $h$ . When the disruption probability is higher than certain threshold,  $\frac{\mu_s^0(h+e-c)}{c\varepsilon}$ , the buyer may choose to the backup as a single source. Note that this threshold decreases further as the correlation factor  $\varepsilon$  increases. The intuition behind this is that when the correlation is higher, the decision maker may consider the reliable as a sole resource even if the primary supplier is not too risky to order.

**Corollary 3** In case of perfect disruption and worthless demand information, if there is no correlation between the disruption state and the spot market price distribution, then the optimal order quantity and optimal reservation capacity are given by

When  $e > c$

- If  $p \leq \frac{h}{\mu_s - e}$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s}\right)$ ,  $k = 0$ .
- If  $\frac{h}{\mu_s - e} < p$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s}\right)$ ,  $k = F^{-1}\left(1 - \frac{h+pe}{p\mu_s}\right)$ .

When  $e \leq c$

If  $p \leq \frac{h+e-c}{\mu_s - c}$ ,  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s}\right)$ ,  $k = 0$ .

If  $\frac{h+e-c}{\mu_s - c} < p$ ,  $q_c = \max\left\{0, F^{-1}\left(1 - \frac{c}{\mu_s}\right) - k\right\}$ ,  $k = F^{-1}\left(1 - \frac{(h+e)-(1-p)c}{p\mu_s}\right)$ .

Corollary 3 corresponds to the results of Proposition 3 when there is no correlation between the spot market price distribution and the supplier's disruption status. An important observation is that in the absence of correlation, the buyer never considers the backup supplier as a sole source of supply.

### 4.3 Worthless information

In this section, we study a setting where demand and supply signals are simultaneously worthless. Since there is no new information in the second stage, execution quantity can be set equal to the reserved capacity at the beginning ( $q_e = k$ ). We may then formulate a newsvendor problem with two suppliers: the reliable (backup) supplier with ordering cost of  $(h + e)$  and the unreliable supplier (with disruption risk of  $p$ ) and with ordering cost of  $c$ . We next state the expected profit for this problem.

$$\pi = r\mu_D - k(h + e) - p\mu_s^1 \Delta(k) - (1 - p)[cq_c + \mu_s^0 \Delta(q_c + k)].$$

Partial derivatives of the objective function with respect to  $q_c$  and  $k$  can be written as  $\frac{\partial \pi}{\partial q_c} = (1 - p)(-c + \mu_s^0(1 - F(q_c + k)))$  and  $\frac{\partial \pi}{\partial k} = -(h + e) + (p\mu_s^1(1 - F(k)) + (1 - p)\mu_s^0(1 - F(q_c + k)))$ .

When the buyer cannot use the option contract to react the new available information, ordering strategy of the firm is determined according to Proposition 4.

**Proposition 4** In case of worthless information, the optimal order quantity and optimal reservation capacity are given by

- If  $p \leq \frac{h+e-c}{\mu_s^1-c}$ , then  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right)$  and  $k = 0$ .
- If  $p \geq \frac{\mu_s^0(h+e-c)}{c(\mu_s^1-\mu_s^0)}$ , then  $k = F^{-1}\left(1 - \frac{h+e}{p\mu_s^1+(1-p)\mu_s^0}\right)$  and  $q_c = 0$ .
- Otherwise  $k = F^{-1}\left(1 - \frac{h+e-(1-p)c}{p\mu_s^1}\right)$ , and  $q_c = \max\left\{F^{-1}\left(1 - \frac{c}{\mu_s^0}\right) - k, 0\right\}$ .

Corollary 4 summarizes the optimality conditions for the setting when the spot market price is independent from the disruption risk.

**Corollary 4** In case of worthless information, if there is no correlation between the disruption state and the spot market price distribution, then the optimal order quantity and optimal reservation capacity are given by

- If  $p \leq \frac{h+e-c}{\mu_s^1-c}$ , then  $q_c = F^{-1}\left(1 - \frac{c}{\mu_s^0}\right)$  and  $k = 0$ .
- Otherwise  $k = F^{-1}\left(1 - \frac{h+e-(1-p)c}{p\mu_s^1}\right)$ , and  $q_c = \max\left\{F^{-1}\left(1 - \frac{c}{\mu_s^0}\right) - k, 0\right\}$ .

Note that the results of Corollary 4 follow the results of Corollary 3 (assuming perfect disruption information) when  $e \leq c$ . According to Corollary 3, when  $e > c$  and the buyer receives the perfect disruption information at stage 2, the buyer's threshold for ordering solely from the primary supplier is  $\frac{h}{\mu_s - e}$ . According to Corollary 4, this threshold is significantly higher for the buyer with worthless information, represented as  $\frac{h+e-c}{\mu_s-c}$ . This result implies that the buyer is more likely not to reserve capacity if the execution cost is high and the buyer cannot take advantage of the flexibility provided by the option contract with information updates.

## 5 Numerical study

In this section, we provide a summary of the results of our numerical study. We specifically examine how the information quality and the level of risk aversion influence the optimal procurement quantities and the corresponding expected profit. The parameter settings used in the numerical study are provided in Table 3. All tests were implemented on an Intel Core™ i5-8250U CPU, 1.60 GHz processor with 8 GB RAM. We utilized CPLEX as a linear programming solver, GAMS as a modelling framework and Java programming language to generate the GAMS model files.

In the numerical study, we assume that the joint distribution of information  $U$  and demand  $D$ , is a bivariate normal distribution with means  $\mu_d$  and  $\mu_u$  and standard deviations  $\sigma_d$  and  $\sigma_u$ , and the correlation coefficient  $\rho$ . We chose problem parameters so that a possibility of negative demand is negligible. The conditional density (density of updated demand)  $f(d|u)$  is normal with the mean  $\mu(u) = \mu_d + \rho \frac{\sigma_d}{\sigma_u}(u - \mu_u)$  and the standard deviation  $\sigma(u) = \sigma_d \sqrt{1 - \rho^2}$ . When  $\rho = 0$ , the second stage information is worthless, and the density of demand at  $t_2$  follows normal distribution with mean  $\mu_d$ , standard deviation  $\sigma_d$ . When  $\rho = 1$ , the information in the second stage is perfect, the demand  $D$  is a deterministic function of  $U$  ( $D|U = \mu_d + \frac{\sigma_d}{\sigma_u}(u - \mu_u)$ ). Notice that  $\sigma(u) \leq \sigma_d$  hence, the bivariate normal distribution lets us to capture how information enables the retailer to obtain a more accurate demand forecast. In this study, we consider cases where  $0 \leq \rho \leq 1$  (increments of 0.1).

The disruption signal in the second stage is assumed to be only  $100\gamma\%$  accurate. For varying situations, given  $\gamma$  and  $p$ , the posterior probabilities are determined, i.e.,  $p(1)$  and  $p(0)$  as described in Sect. 3. We tested cases where  $0.5 \leq \gamma \leq 1$  (increments of 0.1). We also let spot market price to be normally distributed based on the disruption status of the supplier, i.e.,  $N(\mu_s^0, \sigma_s^0)$  for the no-disruption and  $N(\mu_s^1, \sigma_s^1)$  for the disruption setting.

We utilized the scenario-based modelling approach presented in Sect. 3. We employed Monte Carlo simulation to generate second and third stage scenarios based on the distributions described above. Specifically, we generated 200 s stage scenarios with equal probabilities. These scenarios clearly describe the demand signal ( $U$ ) and supply disruption signal ( $I$ ). Using the signals observed in the second stage, we further generated 400 third stage for each second stage scenario, again with equal probabilities. These third stage scenarios clearly describe the customer demand, spot market price and the supplier's disruption status.

For each information setting, we studied the effect of risk aversion by varying the significance level in the range  $0 \leq \alpha \leq 0.9$  (increments of 0.1). The implication of setting the significance level to  $\alpha$  is that we focus on maximizing the expected profit of the smallest  $100 * (1 - \alpha) \%$  scenarios only. For example, when  $\alpha = 0.9$ , we are maximizing the lowest 10% of the outcomes, while when  $\alpha = 0$ , we are maximizing the expected profit. Therefore, as  $\alpha$  increases, the risk aversity increases and the focus is more on the worst-case outcomes rather than the expected outcomes.

We next summarize the results of our numerical tests for various settings. Whenever possible, we will compare the numerical results with the analytical results obtained from the

**Table 3** Problem parameters for the numerical study

| $p$ | $c$ | $e$ | $r$ | $h$ | $\mu_d$ | $\sigma_d$ | $\mu_u$ | $\sigma_u$ | $\mu_s^0$ | $\sigma_s^0$ | $\mu_s^1$ | $\sigma_s^1$ |
|-----|-----|-----|-----|-----|---------|------------|---------|------------|-----------|--------------|-----------|--------------|
| 0.3 | 20  | 21  | 50  | 5   | 2000    | 600        | 1000    | 300        | 35        | 5            | 45        | 10           |

earlier propositions, specifically focusing on the special cases involving risk-neutral decision makers.

## 5.1 Effect of disruption information quality

In this section, we study the effect of disruption information on the procurement portfolio of firms with varying risk attitudes. We provide our analysis on the value of disruption information separately for the settings where the demand information received in stage 2 is perfect ( $\rho = 1$ ) and worthless ( $\rho = 0$ ). We studied settings where  $\gamma$  takes on values of 0.5, 0.6, 0.7, 0.8, 0.9, and 1. Figure 2 illustrates our results for this section.

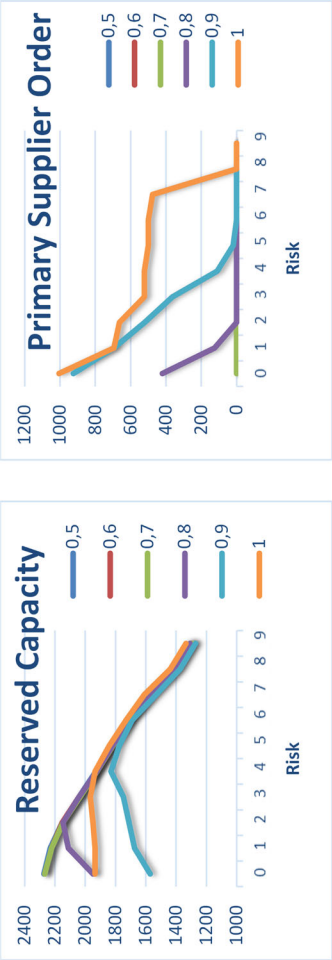
Initially, we evaluate the risk-neutral decision maker's setting ( $\alpha = 0$ ) to validate the analytical findings discussed in Sect. 4. We begin by considering the scenario where both demand and supply information updates are perfect ( $\gamma = \rho = 1$ ). According to Proposition 1, if the disruption probability is below the threshold  $p < -0.18$ , as calculated by the

formula  $\frac{\mu_s^0 \left( \frac{h}{\mu_s^0 - e} \right) - c}{\mu_s^0 \left( \frac{\mu_s^0 - e}{\mu_s^0 - e} \right) - c}$ , it is optimal not to reserve any capacity at the backup. Therefore, for

our numerical setting, the decision maker would consistently choose to reserve some capacity at the backup. Also, as the unit cost for the primary supplier is lower than the execution cost of the option contract ( $c < e$ ), the buyer never prefers to procure solely from the backup supplier. Therefore, in the optimal solution, the firm utilizes both sources. We observe this result from Fig. 2a., when both the demand and disruption information is perfect, a risk neutral decision maker ( $\alpha = 0$ ) orders a substantial quantity from both the primary and the secondary supplier.

We next consider the case where the decision maker has a perfect disruption information ( $\gamma = 1$ ) while the demand information revision is worthless ( $\rho = 1$ ). In this case, we study the risk-neutral decision maker's setting ( $\alpha = 0$ ) to validate the analytical findings discussed in Sect. 4.2. According to Proposition 2, the reservation quantity at the backup cannot exceed  $\theta_1 = F^{-1} \left( 1 - \frac{e}{\mu_s^0} \right) \cong 2050$ . Furthermore, since  $e > c$  and the threshold  $\frac{h}{\mu_s^0 - e} \cong 0.208$  is lower than  $p = 0.3$ , Proposition 3 recommends the following primary order and the reservation capacity values:  $q_c = F^{-1} \left( 1 - \frac{c}{\mu_s^0} \right) \cong 1892$  and  $k = F^{-1} \left( 1 - \frac{h + pe}{p\mu_s^0} \right) \cong 1411$ . The reader can observe similar order quantities from Fig. 2b. Although small variations may occur due to the sampling, the order quantities appear to be consistent with the analytical solutions.

Next, we proceed to evaluate the effect of risk aversion on the procurement portfolio. According to Fig. 2a, when both the demand and disruption information is perfect, a risk neutral decision maker ( $\alpha = 0$ ) orders from the primary supplier around 1000 units and reserves a capacity of around 1900 units at the backup to hedge the disruption risk. On the other hand, a risk averse decision maker ( $\alpha = 0.9$ ) with the same information quality does not order from the primary supplier while keeping a reserved quantity of 1335 units at the backup. Similar findings can be observed in Fig. 2b, for scenarios with worthless demand information. When the risk aversion increases, the order quantity from the primary supplier tends to decrease. Furthermore, beyond a certain level of risk aversion, it is observed that no orders are placed with the primary supplier at all. This indicates that higher levels of risk aversion led to more cautious decision-making, resulting in reduced reliance on the primary supplier. On the other hand, the reservation quantity increases with risk and then decrease again as the decision maker is extremely risk averse. We believe that this is a significant result



a. Perfect Demand Information Setting.



b. Worthless Demand Information Setting.

Fig. 2 Effect of risk aversity and imperfect supply disruption information on the primary supplier order and the reserved capacity at the backup

indicating how supply portfolio of risk neutral and risk averse decision makers deviate when the supplier is subject to a complete disruption and the spot market price is correlated with the disruption.

According to Fig. 2, it can be observed that for a given risk profile, as the quality of disruption information improves, the buyer tends to order higher quantities from the primary supplier. This finding suggests that better information about disruptions enables the buyer to make more informed and confident decisions, leading to increased reliance on the primary supplier. As the quality of the signal for the supplier's status weakens, the buyer becomes increasingly hesitant towards relying solely on this supplier. In such cases, the buyer may exhibit a preference for reserving capacity at the backup to mitigate the potential risks associated with disruptions related to the primary supplier.

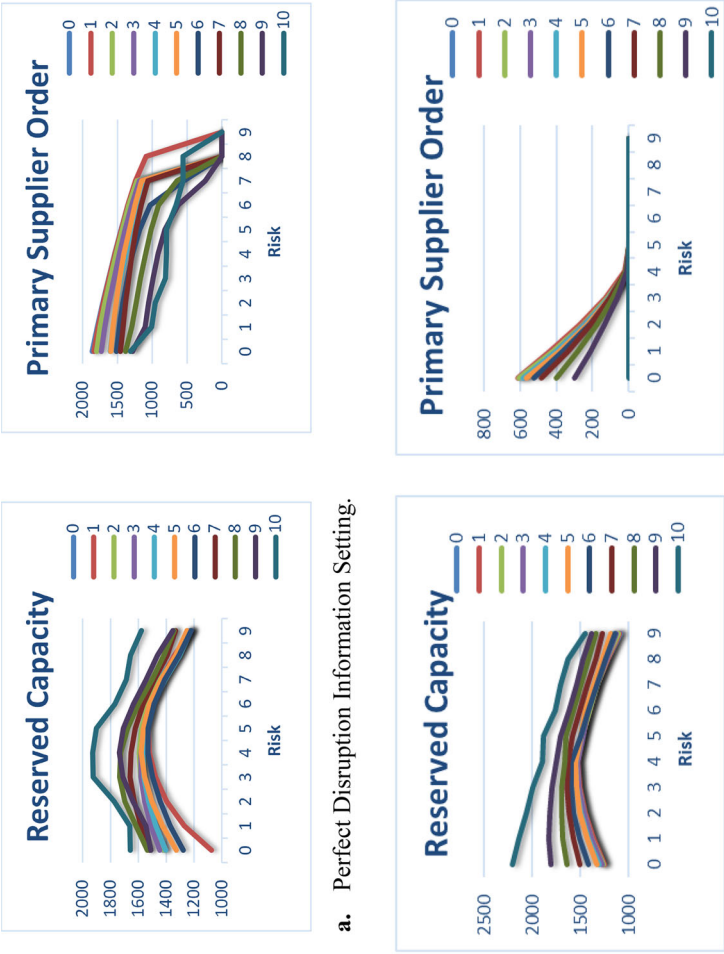
Another key result from Fig. 2 is that the effect of disruption information is more evident for a risk neutral decision maker compared to a risk neutral decision maker for both perfect and worthless demand information settings. For instance, with perfect demand information; a risk averse decision maker ( $\alpha = 0.9$ ) orders nothing from the primary supplier and reserves approximately 1300 units at the backup supplier regardless of the quality of disruption signal. A similar pattern is observed for a decision maker with worthless demand information. This indicates that the decision maker prioritizes risk mitigation and relies heavily on the backup supplier to ensure a stable supply for varying levels of information. In contrast, the procurement mix of a risk-neutral decision maker varies significantly with varying levels of the disruption information and demand information. A risk-neutral decision maker is therefore more responsive to the quality of information.

## 5.2 Effect of demand information quality

Next, we assume that the buyer receives imperfect demand information in stage 2. We specifically evaluate cases where  $\rho$  ranges from 0 to 1, with increments of 0.1. For each imperfect demand information scenario, we consider two settings: one with perfect supply disruption information and another with worthless supply disruption information. Figure 3 provides a summary of our results for these specific settings, offering insights into the impact of imperfect demand information on the supply portfolio in combination with different levels of supply disruption information.

We first study the risk neutral decision maker's setting ( $\alpha = 0$ ) to validate our analytical results for the case where both demand and supply information updates are worthless ( $\gamma = \rho = 0$ ). For this specific numerical setting,  $\frac{h+e-c}{\mu_s^1-c} = 0.24 < p = 0.3 < \frac{\mu_s^0(h+e-c)}{c(\mu_s^1-\mu_s^0)} = 1.05$ . Proposition 4 then suggests reserving a quantity of  $k = F^{-1}\left(1 - \frac{h+e-(1-p)c}{p\mu_s^1}\right) \cong 1267.616$  units at the backup supplier, while placing an order of  $q_c = \max\left\{F^{-1}\left(1 - \frac{c}{\mu_s}\right) - k, 0\right\} \cong 692$  units with the primary supplier. It is noteworthy that similar quantities can be observed in Fig. 3b, providing empirical validation of the analytical findings.

Consistent with our earlier findings, Fig. 3 supports the observation that as the firm aims to minimize risk, the utilization of the unreliable supplier significantly decreases. This indicates that when risk aversion is high, the decision maker becomes increasingly hesitant to rely on the unreliable supplier as a viable supply source. It is interesting to note that when the decision maker has perfect supply disruption information, they are more likely to continue ordering some quantity from the primary supplier, despite their risk aversion. This suggests that the decision maker recognizes the potential benefits of maintaining a certain level of supply from



**Fig. 3** Effect of risk aversity and imperfect demand information on the primary supplier order and the reserved capacity at the backup

the primary supplier, even in the presence of risk. However, in situations where the decision maker exhibits extreme risk aversion, the firm completely disregards the unreliable supplier as a viable supply source, regardless of the quality of disruption information. Instead, the firm prefers the backup supplier, to ensure a more secure supply.

Finally, according to Fig. 3, improved demand information quality leads to increased capacity reservation through option contracts and decreased reliance on the unreliable supplier. This is because better demand information allows the buyer to utilize backup capacity more efficiently. This effect is observed for both risk-neutral and risk averse decision makers.

### 5.3 Relative value of information for a risk averse decision maker

We next evaluate the expected (monetary) value of demand and disruption information for decision makers with varying risk considerations. Figure 4 illustrates the average of expected profit values over five imperfect disruption information levels for both the perfect and worthless demand information settings. Based on this figure, we can conclude that the expected value of demand information for a risk averse decision maker is negligible compared to a risk-neutral decision maker. A similar comparison was conducted for the value of disruption information. According to Fig. 5, our conclusion holds true in this case as well, indicating that the expected value of disruption information for a risk averse decision maker is significantly smaller compared to a risk-neutral decision maker.



Fig. 4 Average expected profit values across five levels of imperfect disruption information



Fig. 5 Average expected profit values across eleven levels of imperfect demand information

## 6 Summary of key results

This section summarizes key insights and managerial implications derived from the analytical results and computational tests.

In the risk-neutral setting, we develop optimality conditions for various special cases based on the information quality when the firm utilizes the reserved capacity at the backup. In addition to characterizing the exact ordering quantity at the unreliable supplier and the reservation capacity at the backup supplier, we provide threshold values for the disruption risk describing firm's preference towards to single or multi-sourcing strategies. We find out that when there is no correlation, a risk neutral buyer never orders exclusively from the backup supplier as the backup supplier is not cost efficient. This result is known as “cost first, reliability second” in the unreliable supplier selection literature. We demonstrate that increased correlation between spot market price and the disruption event enhances the attractiveness of the backup supplier. In such cases, the flexible but not efficient backup supplier becomes preferable and can be chosen as a single source. We believe that this is a significant result, contributing to the risk neutral procurement literature, as it highlights the crucial role of the correlation between supply disruptions and the spot market price. Similarly, we show that it is optimal to reserve more capacity at the backup supplier as the quality of the disruption signal improves. We also validate our analytical findings comparing them with the numerical outputs obtained through our scenario-based modelling approach for the risk neutral decision maker.

For risk averse firms, increasing quality of supply disruption information and demand information make the backup supplier more valuable, as the reserved capacity can be more efficiently utilized in such cases. As the quality of disruption information improves, the buyer tends to order higher quantities from both the primary supplier and the backup supplier. On the other hand, as the quality of demand information improves, the buyer tends to decrease the orders from the primary supplier and shift their orders to the backup supplier.

For extremely risk averse decision makers, the firm exhibits a strong tendency to completely disregard the unreliable supplier as a sustainable supply source, irrespective of the quality of disruption information. Hence, the anticipated impact of demand and disruption information is reduced for risk averse decision makers compared to their risk-neutral counterparts. This leads to a reduced expected monetary value of demand and disruption information for decision makers who are risk averse.

## 7 Implications on sustainable development goals

Our research supports the broader agenda of achieving the SDGs by 2030. In this section, we summarize our contributions in this respect.

Our study provides practical decision models that can assist organizations in navigating uncertain environments and making well-informed decisions. By considering elements such as supply disruptions, volatile spot market prices, and option contracts, our models can help organizations optimize their procurement processes, mitigate risks, and enhance overall operational efficiency. Therefore, our research aligns well with SDG 9 (Industry, Innovation, and Infrastructure).

By highlighting the role of flexible options and correlation between supply disruptions and spot market prices, we minimize possible overproduction (waste) and under production (shortages). This would assist responsible procurement practices from both environmental

and the economic perspectives. Therefore, our research contributes to SDG 12 (Responsible Consumption and Production).

Our research can be applied to settings to address humanitarian supply chain challenges contributing SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities). Ensuring a reliable supply of urgent items during public health emergencies or natural disasters such as pandemics or earthquakes is vital to minimize losses in the affected populations. By leveraging risk averse, multi-stage stochastic programming models with real-time updates on demand and supplier disruptions, our models can empower non-profit organizations to optimize their procurement portfolios, and to accelerate the effective response and fairness.

## 8 Conclusion

This study aims to improve the state of knowledge on the impact of forecast updating in a supply chain, as firms generally face a trade-off between the cost and the benefit of gathering market and supply information. To the best of our knowledge, in the existing supply chain literature, the focus is generally on value of demand information for risk neutral decision makers. In our study, we simultaneously focus on the value of supply disruption information and demand information. Moreover, we incorporated risk aversion into our analysis. Furthermore, we study settings where the distribution of the random spot market price depends on the unreliable supplier's disruption status. Although the spot market price is often affected by supply disruptions in practice, this is not a common assumption in most of the cited literature.

Despite the inherent complexity of the decision models, this study provides analytical solutions for several special cases considering the risk neutral decision maker. Optimality properties lead to notable insights on the ordering strategies for firms. For example, we show that the traditional “cost first, reliability second” result in the unreliable supplier selection literature does not hold when there is correlation between the spot market price and the supplier disruption. We believe that this is a significant result, contributing to the risk neutral procurement literature, as it highlights the crucial role of the correlation between supply disruptions and the spot market price.

Moreover, we carry out an extensive numerical study to assess the value of information and the effect of risk aversion on procurement decisions. For example, we establish that the expected value of information is higher for the risk neutral decision maker compared to the risk averse decision maker. This is a fundamental result for decision makers who are evaluating options to collect more information before using the flexible resource. Finally, the consistency between the analytical results and the numerical tests strengthens the robustness and reliability of the study's conclusions.

By exploring procurement strategies that consider risk attitudes and information quality, this study promotes sustainable supply chain management practices. Therefore, in addition to its contributions to supply chain decision-making literature, this study aligns well with the objectives of the Sustainable Development Goals (SDGs).

In this study, we examine the value of information for the risk averse decision maker with the CVaR objective. A future study may also consider alternative approaches for the risk averse models. Utilizing a stochastic programming approach with uncertain disruption scenarios raises a practical concern that recourse decisions may deviate significantly from the corresponding pre-determined first stage decisions, increasing system nervousness. In this respect, future studies could focus on developing models with restricted recourse, for example employing CVaR norm (see Pavlikov & Uryasev, 2014; Koca et al., 2021). We

assumed that spot market price and the disruption status of a single supplier are correlated. We therefore assume that customer demand is independent from supply uncertainties. Future research can analyze scenarios where multiple suppliers exist, and demand is correlated with supply chain disruptions and the spot market prices. Moreover, it would be interesting to consider settings where the firm's initial prediction on supply disruption is biased.

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